Race, explain and transfer

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Evaluate a model against measured performance and defend design decisions.

SUCCESS CRITERIA - I can...

1. Report measured distance, time and average speed.
2. Compare actual speed with a stated prediction.
3. Use evidence to explain three physical design choices.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “relative deviation” and “energy budget” to explain your model or result. Sentence starters are optional.

relative deviation: difference from prediction compared with its magnitude

energy budget: an account of all energy stored and transferred

revision: a change made after checking evidence or reasoning

01 Warm-up: bring back an earlier idea

A 60-tooth driver turns a 480-tooth gear.
Give ideal output/input torque and speed ratios and state the loss assumption.

Word help: torque = the turning effect of a force about an axis

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: ① ② ③

02 Worked example: explain the model

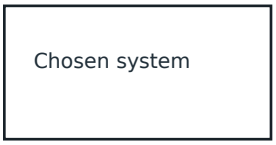
Reference: average speed = distance/time
relative deviation = |measured - predicted|/|predicted| × 100%. Prediction must be nonzero. Record real race conditions; this compares with a prediction, not symmetric percent difference.

Worked reasoning: A car covers 3 m in 2 s: average speed 1.5 m/s. A complete explanation also uses its circuit, gearing, friction and evidence from repeated runs; one race time does not isolate a cause.

Your turn: Draw an energy-flow diagram from battery to motion and other transfers; annotate it with one actual result, clearly labeled as measured.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Draw and label all energy transfers across the boundary.



Stored energy change: _____ J

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Draw or annotate your final car.
Label three choices involving wheels, gears, alignment, wiring or mass, then connect each to a physics idea.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Marzano check - circle: 1 2 3

04 Apply the physics

A supplied car covers 6.0 m in 8.0 s; its predicted speed was 0.90 m/s. Complete the table, find measured speed and percent deviation from prediction.
Give a plausible explanation and identify a measurement that could test it.

Sentence starter: I predict ... . My model supports this because ... .

Distance (m)	Time (s)	Measured v	Predicted v
___	___	___	___

Marzano check - circle: 1 2 3

05 Explain your reasoning

Use your actual race record to defend one successful design choice and identify one revision for a future build. Cite data or a specific observation for each.

Sentence starter: My design shows ... . It meets the requirement by ... .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A motor is 64% efficient and the following generator is 72% efficient.
Find the combined useful fraction in sequence.
Label input, useful output and other energy transfers.

Word help: generator = a device converting mechanical input to electrical output

Sentence starter: I found ... [unit]. This value means ... in this situation.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: 1 2 3

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Continue down each column.

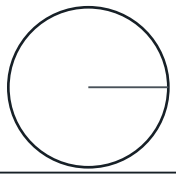
Sentence starters are optional. You may plan in any language.

07 Exit check

Transfer: a similar car uses twice the wheel diameter with the same loaded wheel rpm.
Predict ideal speed change, then explain why testing is still necessary.

Word help: ideal = using stated simplifying conditions

Sentence starter: I predict … . My model supports this because … .



Radius / diameter: ____

Turns per second: ____

Travel per turn: ____ Ground speed: ____

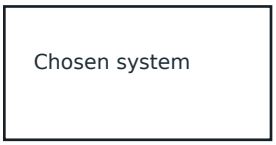
Marzano check - circle: 1 2 3

08 STRETCH: test the model

Supplied input is 6 V×1 A for 4 s. A 0.5 kg car reaches 2 m/s on a level track from rest.
Find input energy and final translational kinetic energy.
Explain why their ratio alone is not total drivetrain efficiency.

Sentence starter: Under the assumption …, the evidence … supports … because … .

Draw and label all energy transfers across the boundary.



Stored energy change: _____ J

Marzano check - circle: 1 2 3